

Foundation (Jalapeno)

- Q1.** What is the greatest common factor of $6x$ and $9x$?
(A) $3x$
(B) 6
(C) 9
(D) 12
(E) 18
- Q2.** Factor out the greatest common factor from $8x + 12$
(A) $4(2x + 3)$
(B) $2(4x + 6)$
(C) $8(x + 1)$
(D) $12(x + 2)$
(E) $16(x + 3)$
- Q3.** What is the greatest common factor of 20 and 30?
(A) 5
(B) 10
(C) 15
(D) 20
(E) 25
- Q4.** Simplify the expression $9x^2 + 12x$ by factoring
(A) $3x(3x + 4)$
(B) $3x(3x + 2)$
(C) $6x(3x + 2)$
(D) $9x(x + 1)$
(E) $12x(x + 2)$
- Q5.** Factor out the greatest common factor from $15x^2 + 25x$
(A) $5x(3x + 5)$
(B) $5x(3x + 2)$
(C) $10x(3x + 2)$
(D) $15x(x + 1)$
(E) $20x(x + 2)$
- Q6.** What is the greatest common factor of 24 and 36?
(A) 6
(B) 8
(C) 12
(D) 18
(E) 24
- Q7.** Simplify the expression $16x + 20$ by factoring
(A) $4(4x + 5)$
(B) $4(4x + 2)$
(C) $8(2x + 5)$
(D) $10(2x + 1)$
(E) $16(x + 2)$
- Q8.** Factor out the greatest common factor from $12x^3 + 15x^2$
(A) $3x^2(4x + 5)$
(B) $3x^2(4x + 2)$
(C) $6x^2(2x + 5)$
(D) $9x^2(x + 1)$
(E) $12x^2(x + 2)$

Open Ended Questions (FRQ)

- Q9.** Factor out the greatest common factor from $18x^2 + 24x$. Show all steps and simplify.
- Q10.** Simplify the expression $25x + 30$ by factoring. Show all steps and simplify.

Chef's Tips

- Tip 1: Make sure students show all steps when simplifying expressions.
- Tip 2: Emphasize the importance of factoring out the greatest common factor.
- Tip 3: Encourage students to check their work by multiplying the factors